

Shaun Allison

shallison@protonmail.com • <https://shalliso.github.io/> • <https://github.com/shalliso>
Toronto, Canada • US citizen and Canadian permanent resident

Software engineer and logician with experience in both industry and academia. After almost a decade of doing pure mathematics research I'm returning to industry. Good fits for me include formal verification, AI research, quantitative research, data science, and software engineering and my colleagues have gone on to enter all of these fields.

SKILLS

- **Languages:** Python, Lean (current); Hack, JavaScript, Haskell, C (industry use, 2012–2016)
- **Tools:** git, $\text{\LaTeX}$; familiar with numpy, matplotlib, PyTorch
- **Domains:** formal verification, machine-learned ranking and targeting, mathematical logic

EXPERIENCE

Software Engineer — Meta June 2015 – August 2016

- Used machine learning to improve targeting of advertisements on Facebook Games, including notifications and image assets.

Postdoctoral Fellow / Lecturer — Univ. of Toronto; Hebrew Univ. of Jerusalem 2021 – 2025

- Independent research program in mathematical logic; published in the *Journal of the European Mathematical Society*, *Advances in Mathematics*, and the *Journal of Mathematical Logic*.
- Instructor of record for eight courses, including a 400-student calculus class — scoping a syllabus, building assessments, and managing teaching assistants.

Independent Research January 2026 – present

- Two Lean 4 formalization projects (below); carried two journal papers through revision to acceptance.
- Writing a textbook on mathematical logic with Spencer Unger, alongside continuing research in descriptive set theory.
- Research visits to Cornell University and the University of Vienna; upcoming visit to the Institute for Advanced Study in Mathematics, Zhejiang University.

Earlier: software engineering internship at Meta (2014); research and development at VMware (2013); Cisco (2012).

PROJECTS

Knight model formalization — Lean 4, Mathlib, 2026 <https://github.com/shalliso/KnightModel>

- ~5,700 lines of Lean formalizing a generalization of a classical model-theoretic construction. Developed with AI assistance (Claude) under my direction.

Automatic continuity — Lean 4, Mathlib, 2025 https://github.com/shalliso/automatic_continuity

- ~300 lines of Lean, written by hand. Complete machine-checked proof of a classical theorem in topological group theory.

PATENTS

- *Automatic recipient targeting for notifications* (US10506055B2) — machine-learned selection of notification recipients. Granted.
- *Selecting assets* (US11019177B2) — machine-learned selection of which creative asset to serve. Granted.

EDUCATION

- **Ph.D. Mathematics**, Carnegie Mellon University, 2021
- **B.S. Computer Science** and **B.S. Mathematics** (Discrete Mathematics and Logic), Carnegie Mellon University, 2015
- Putnam Competition top 500; CMU College Honors (Computer Science); Phi Beta Kappa